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Way to solve power problems in developing countries

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MANY DEVELOPING COUNTRIES LACK ACCESS TO RELIABLE ENERGY

1.0 INTRODUCTION

In many developing nations, access to reliable and affordable electricity remains a challenge. Millions of humans continue to depend on conservative energy sources such as firewood and kerosene causing inefficient, health problems and environmental hazards. This dependence on old-school methods drags progress in education, business, and quality of life. The underlying issues fueling this energy crisis are present from limited modern technological solutions and a shortage of professionals in implementing and maintaining such systems. Until then, fair access to modern energy will remain out of reach for many communities.

This event is inciting many developing areas to find themselves trapped in a four-wall of energy poverty. It is really difficult to access crucial services like healthcare, education, and job opportunities due to the decrease of reliable electricity. Long-term health problems and harmed environments occur when we depend on unstable and temporary energy sources.

Pristine, secure and lasting energy solutions are needed to break this cycle. Thanks to advanced technologies, renewable energy is becoming more accessible, eco-friendly, and altered to meet the needs of rural societies. This report discovers three practical renewable energy options that can resolve these issues such as biomass gasification, geothermal, and decentralized energy. These solutions have the potential to lead to sustainable development, decreasing our reliance on dangerous energy sources, and open a new door to economic growth.

2.0 BACKGROUND OF THE PROBLEM

The terrible energy crisis in developing countries is the primary problem caused by the two related issues which are a lack of appropriate technology and an insufficient skilled workforce. Without enough access to any distribution technologies and advanced energy, these nations struggle to initiate and maintain the power grids. Furthermore, even when the technology already occurs in this era, the absence of trained engineers, technicians and maintenance employees means that the infrastructure often fails, wasted or quickly falls into abandonment. This technological and shortage of quality employees directly impacts a heavy reliance on harmful energy sources like using the firewood can lead to health especially respiratory problems and environment problems. These ways also lead to illnesses and increase greenhouse gas emissions

where over 75 percent of global greenhouse gas emissions come from it (*Causes and Effects of Climate Change*, n.d.). All of These harmful methods keep people continuing to harm the environment and it can make it worse.

3.0 POSSIBLE SOLUTIONS

Addressing the energy shortage in developing countries requires a clear strategy that uses a variety of renewable energy sources. The following are possible solutions to be discussed.

3.1 Biomass Gasification

Biomass gasification is an exciting renewable energy solution that transforms organic waste—like agricultural leftovers, wood chips, and animal manure—into power through a high-temperature chemical process (Sikarwar et al., 2016). This method is particularly beneficial for developing countries, especially in communities that rely heavily on farming and livestock, where agricultural waste is abundant and often discarded. By setting up small to medium-sized gasification units near farms, cooperatives or local governments can turn this waste into energy for small-scale industries, cooking, lighting, and irrigation. The advantages are significant, including low fuel costs, reduced environmental waste, and the ability to operate off-grid, making it an ideal choice for remote areas with limited infrastructure while also supporting a circular economy. However there are some challenges to consider, like the need for regular maintenance, a reliable source of dry biomass, and skilled workers to keep everything running smoothly (Baruah & Baruah, 2014). If gasifiers aren't properly looked after, they could emit harmful substances like tar and particulate matter.

3.2 Geothermal Energy

Geothermal energy is a fantastic renewable energy option for developing countries. It taps into the heat stored beneath the Earth's surface to generate electricity or provide direct heating (IRENA, n.d.). This practice is especially effective in areas near volcanic mountains, where the heat sources are closer to the surface. To harness this energy, deep wells are drilled to access geothermal reserves, and power plants are built to supply local communities with a steady stream of electricity. Geothermal energy delivers a reliable baseload power supply 24/7, which is one of its biggest advantages. Plus, once

it's up and running, it produces relatively low greenhouse gas emissions, making it a green choice for the environment. Well, there are some challenges to consider—like the need for specific geological conditions, the hefty initial costs of exploration and drilling, and potential risks such as ground instability or the release of trace gases (Barbier, 2002).

3.3 Decentralized Energy System

Decentralized Energy Systems provide a new better way of delivering the renewable energy, switching from large, centralized then smaller systems. These systems usually generate and supply power directly to the nearby areas and rural areas (Anand et al., 2021). The biggest advantage of developing this energy system is it can bring clean and dependable energy to rural areas that are not accessible and not connected to the main power grid. It can be easily deployed by anyone willing to learn how to install the system in their own areas. The energy system also provides enough power and energy to those who need it the most without the high cost of building the national power grid infrastructure (Ulsrud et al., 2011). However, the system faces financial limits, including to ensure the sustainability of business models and maintaining equipment in remote locations. Even with these issues occurring, the adaptability and benefits of decentralized systems make it a more important part of solving the lack of energy problems in developing countries because they also can gain financial help from the non-governmental organization to deploy this system.

4.0 CONCLUSION

To conclude, the current issue which is energy shortage in developing countries is mainly caused by a lack of modern technology and skilled employees. This will force many communities to only rely on harmful energy sources which can continue to harm both people and the environment. To prevent this from happening, many renewable energy technologies offer effective solutions. By strategically implementing the solutions like biomass gasification in agricultural areas, making use of geothermal energy where available and deploying decentralized energy systems for remote areas play an important role in overcoming their power shortage. Achieving this requires training a local workforce to support and sustain this growth. In the long term, providing energy access to enhance public health, improve education and create new economic opportunities allowing entire communities to achieve sustainable development.

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